Stephen Blandford

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Education

The University of Texas at Austin

- **PhD in Operations Research and Industrial Engineering** May 2025 – Present
  - Selected Coursework: Markov Decision Processes, Deep Learning Methodology and Applications
- **MS in Operations Research and Industrial Engineering** Aug 2023 – May 2025
  - Selected Coursework: Decision Analysis, Stochastic Processes, Energy System Optimization, Microeconomics
- **BS in Mechanical Engineering**, with High Honors Aug 2019 – May 2023

Research Experience

Lawrence Livermore National Laboratory Livermore, CA
Cyber and Infrastructure Resilience Intern May 2026 – Present

- Develop defender–attacker–defender optimization models to identify line hardening decisions that minimize worst-case load shedding under targeted attacks in distribution systems with widespread adoption of distributed generation and islanding capabilities
- Implement single- and three-phase LinDistFlow formulations and an iterative solution algorithm in Python using Pyomo and Gurobi, applying LP duality and mixed-integer reformulations to solve multilevel grid hardening and interdiction problems
- Evaluate how power flow approximations and phase imbalance affect attack and hardening decisions using synthetic distribution feeder models and LLNL's high-performance computing resources
- Formulate and compare path-based and artificial-flow connectivity constraints to restrict grid-following generation to islands with grid-forming sources, while preserving linearity of the operations subproblem

The University of Texas at Austin Austin, TX
Graduate Research Assistant, Advisor: Benjamin D. Leibowicz Jan 2024 – Present

Modeling the Economic Benefits of Grid Hardening Investments Mar 2025 – Present

- Collaborate with multiple institutions to develop an integrated modeling framework for quantifying the economic benefits of power system resilience investments
- Employ a computable general equilibrium (CGE) model in GAMS to translate power availability into economy-wide impacts
- Develop realistic transmission-distribution network models for U.S. regions using validated test systems and public datasets
- Implement joint transmission-distribution repair heuristics and power flow models in Python, integrating LPAC and three-phase LinDistFlow formulations to link grid damage and restoration actions to power availability
- Develop a coastal southeastern U.S. case study evaluating the economic benefits of power line undergrounding by coupling probabilistic hurricane and damage models with grid restoration modeling to assess long-term outage risk

Optimal Joint Transmission-Distribution Repair Jan 2024 – Mar 2025

- Developed and implemented a MILP framework in Python/Pyomo for joint transmission and distribution repair scheduling
- Introduced methods to represent distribution system damage and repair without full component-level data and incorporate resident evacuation and return dynamics into restoration optimization
- Built and calibrated a Florida transmission network model using GIS and publicly available data, and fit fragility models to simulate spatially varying hurricane damage with Hazus
- Demonstrated the framework through a Hurricane Irma case study and conducted sensitivity analyses on repair parameters and critical load weighting to assess robustness and restoration trade-offs

Publications

Stephen K. Blandford and Benjamin D. Leibowicz, 2026. Optimal transmission system repair considering distribution system damage and evacuations. *Reliability Engineering & System Safety* 272, 112568. https://doi.org/10.1016/j.ress.2026.112568 (peer-reviewed)

Presentations

Optimal Transmission System Repair Considering Distribution System Damage and Evacuations Nov 2026
INFORMS Annual Meeting, San Francisco, CA (scheduled)

Teaching

Freshman Introduction to Research in Engineering Program, *Graduate Teaching Fellow* – UT Austin Fall 2025

- Mentored a team of three first-year undergraduate engineering students on an applied power grid restoration research project, guiding their first academic research experience from problem formulation through technical communication
- Taught foundational operations research methods, including linear & integer programming, through collaborative applied examples
- Guided students in developing, coding, testing, and evaluating heuristic repair strategies in Python using a synthetic power grid model, culminating in a 1st-place finish in the program's final presentation competition

Awards

- 1st Place Team, Cybersecurity Capture the Flag Competition – LLNL CSI Summer 2026
- Thrust 2000 Graduate Fellowships in Engineering Fall 2023 – Present
- T. W. Whaley, Jr. Friends of Alec Scholarship Fall 2023
- A.O. Fletcher Endowed Presidential Scholarship Fall 2022 – Spring 2023
- Susan and Robert Campbell Jr. Endowed Scholarship in Mechanical Engineering Fall 2021 – Spring 2023

Service, Leadership, & Memberships

The University of Texas at Austin

INFORMS Student Chapter

Treasurer Fall 2026 – Present
Secretary Fall 2025 – Spring 2026
President Fall 2024 – Spring 2025

Mechanical Engineering Graduate Student Board Fall 2024 – Present
Board Member

Operations Research and Industrial Engineering Peer Mentorship Program Fall 2024 – Present
Mentor

Institute for Operations Research and the Management Sciences (INFORMS)
Member Fall 2024 – Present

Macro-Energy Systems Society
Member Fall 2024 – Present

Skills

Programming & Software: Python (Pyomo, Gurobi), Slurm/HPC, GAMS, MATLAB, GIS, Hazus, Arena Simulation, LaTeX, SOLIDWORKS
Methods: Infrastructure Modeling, Optimization (LP, MILP, NLP), Simulation, Stochastic Processes, Decision Analysis